

Response to Reviewer #3

General Response: We sincerely thank the reviewer for their careful reading and constructive feedback. The suggestions regarding model heterogeneity, non-stationarity, and statistical robustness have been immensely helpful in refining the scope and clarity of our manuscript. In our revised submission, we have significantly expanded the methodology and datasets, directly addressing the core of your concerns while keeping the focus strictly on the comparative advantage of dynamic model selection.

1. Formal Tests for Non-Stationarity

Reviewer’s Comment: *The assertion that the proposed method adapts to non-stationary environments would be strengthened by incorporating formal unit-root or structural-break tests (e.g., ADF, KPSS, or Sup-F).*

Response: We appreciate the reviewer’s econometrics perspective. While formal statistical pre-tests like ADF or KPSS are standard in classical time-series analysis, our deep learning framework is evaluated across a continuous, 24-year out-of-sample period (2000–2024) spanning five highly diverse asset classes. This massive evaluation horizon inherently and demonstrably contains profound macroeconomic structural breaks (e.g., the 2008 Global Financial Crisis, the 2020 COVID-19 crash, and the subsequent high-inflation regime).

Rather than relying on rolling classical pre-tests to definitively prove non-stationarity, we demonstrate the model’s adaptability empirically. As evidenced by the long-term performance trajectories in **Figure 5** and the risk-adjusted metrics in **Table 1**, the MAB successfully adapts to these well-documented historical regime shifts, providing downside protection when Long-Only benchmarks collapse. Furthermore, to rigorously ensure our model is adapting to actual structural breaks rather than just autocorrelated noise, we have added a new **Section 4.2 (Robustness Check: Synthetic Data Simulation)** featuring an EGARCH(1,1) placebo test. We believe this broad, empirical multi-asset evaluation sufficiently validates the strategy’s robustness to real-world non-stationarity.

2. Heterogeneous Set of Base Models

Reviewer’s Comment: *Given that all arms share the same architecture and hyper-parameters, their prediction errors are likely to be highly correlated, which may diminish the exploratory benefits of the bandit framework. A comparative analysis using a heterogeneous set of models such as MLPs, LSTMs, CNNs would better illustrate the added value of the CMAB strategy.*

Response: We completely agree with the reviewer that limiting the pool to identically parameterized MLPs artificially restricts the power of the MAB framework. During our revisions, we entirely overhauled our experimental setup to address exactly this issue.

As detailed in the heavily revised **Section 3.2.1 (Impact of Architectural Heterogeneity)**, we have introduced a heterogeneous predictor pool consisting of MLPs, CNNs, and RNNs. Our new results (now expanded across five different asset classes in **Table 1**) perfectly align with your hypothesis: while introducing this architectural variance actually penalizes the static ensemble average (due to conflicting inductive biases and noise), the explicitly exploratory CMAB policies (such as LinUCB) thrive, effectively turning the varied predictions into a strategic, regime-dependent advantage. We thank the reviewer for this excellent suggestion, as it became a cornerstone of our revised manuscript.

3. Look-Ahead Bias in Context Vectors

Reviewer’s Comment: *The use of context vectors derived from overlapping walk-forward windows could introduce look-ahead bias. To mitigate this, it is advisable to retrain the bandit model from scratch within each window or to adopt a block-bootstrap procedure that disrupts temporal autocorrelation.*

Response: We wish to clarify our pipeline to assure the reviewer that our methodology already implements the exact safeguard suggested. Specifically, for each overlapping walk-forward rolling window, we completely reinitialize and retrain both the base deep learning predictors and the MAB policies from scratch.

Furthermore, all reported performance metrics (e.g., in Table 1 and Figure 5) are computed exclusively on strictly out-of-sample test slices that neither the models nor the bandit have ever observed during their respective training or tuning phases. Because the entire system is reinitialized per window and evaluated strictly forward in time, our walk-forward testing naturally maintains strict temporal separation, rendering any look-ahead bias impossible.

4. Transaction Costs

Reviewer’s Comment: *The current evaluation does not account for transaction costs. To assess practical applicability, it would be informative to report cost-adjusted Sharpe ratios under various one-way fee assumptions.*

Response: We acknowledge that transaction costs are critical for evaluating a production-ready trading strategy. However, the explicit scope and objective of this paper is to isolate and compare the *pure predictive accuracy* and *model selection capabilities* of dynamic CMABs versus static ensemble averaging.

Because CMABs intentionally switch between predictors to adapt to shifting regimes, they naturally incur higher turnover than a static average. Introducing transaction costs at this stage would conflate predictive accuracy with execution efficiency, obscuring the primary theoretical contribution of the paper. We believe that incorporating a proportional transaction cost penalty directly into the RL reward function is an excellent direction for extending this work, and we have explicitly highlighted this in the **Conclusion (Section 5)**.

5. Statistical Robustness and Random Seeds

Reviewer’s Comment: *The statistical robustness of the results is limited by the small number of random seeds. Increasing the seed count and conducting paired statistical tests with appropriate multiple-testing corrections would enhance the reliability of the inferred significance.*

Response: We appreciate the reviewer’s push for statistical rigor. However, we would like to clarify the extensive computational scope of our current experiments. Our evaluation requires training a pool of roughly 60 deep neural networks, and then simulating the sequential bandit selection across more than 6,000 daily time steps.

For the revised manuscript, we conducted 25 fully independent macro-runs for each configuration. Given that we have now expanded the paper to test 3 distinct MAB policies across 5 entirely different asset classes (Equities, Crypto, FX, Commodities, Bonds) using both homogeneous and heterogeneous pools, this amounts to hundreds of distinct macro-simulations and the training of hundreds of neural networks. Given the heavy computational burden of deep learning ensembles, 25 runs per configuration across 5 diverse market microstructures provides highly robust, statistically stable proof of the MAB framework’s comparative advantage over the static ensemble.